

Spectral Pathfinding: Using A* to Navigate Chromophore Networks for FRET Design

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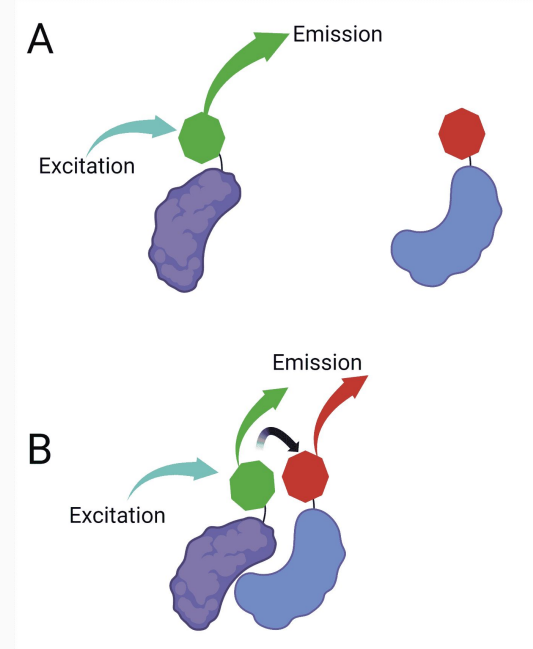
CHEM 274B Software Engineering Fundamentals for Molecular Science

The Problem: FRET Pathway Design

GOAL: Find the optimal multi-step FRET chains

**Natural chlorophyll database for varying spectras
with overlap**

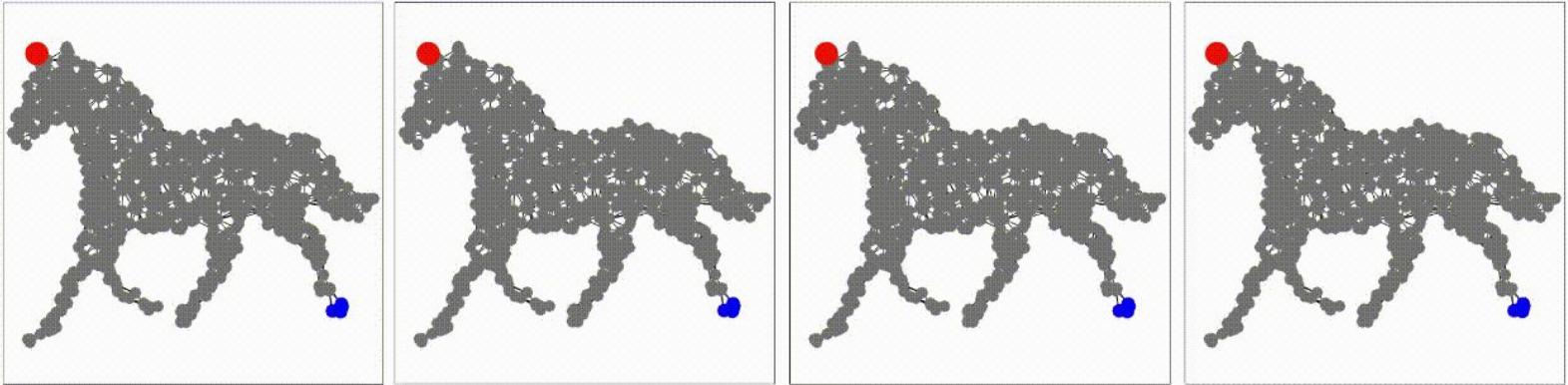
**FRET defines how energy hops, and A* finds the
optimal route those hops should take**



Utilizing A* Search for FRET Pathways

The A* Search Algorithm Visualized

Heuristic Strength →



Utilizing A* Search for FRET Pathways

Dijkstra's Algorithm

- Selects next node based on lowest accumulated cost from start node
- Explores uniformly in all directions
 - Will explore many irrelevant nodes

$$f(n) = g(n)$$

A* Search Algorithm

- Adds heuristic cost to target to the accumulated cost
- Heuristic directs search towards the target node
 - More efficient on average

$$f(n) = g(n) + h(n)$$

	Dijkstra's	A*
Time	$O(E \log V)$	$O(E \log V)$
Space	$O(V)$	$O(V)$

$$\text{A* Effective Time} = O(b^d)$$

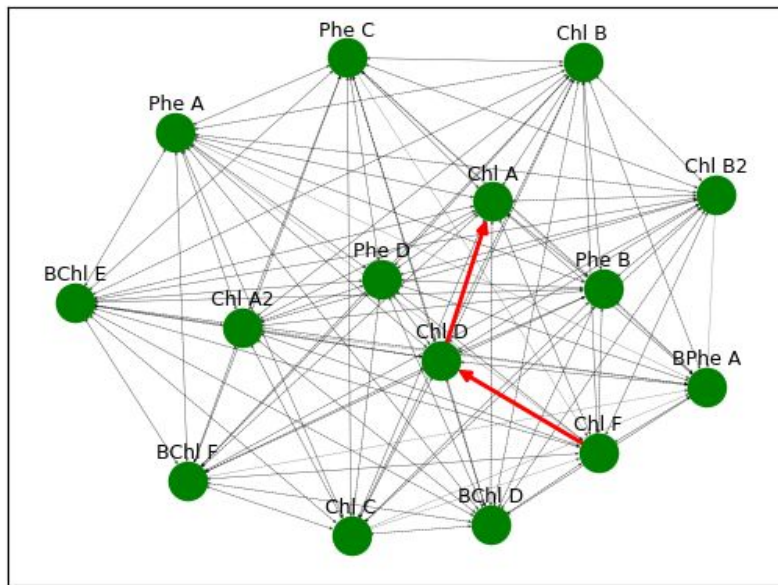
b = branching factor

- Avg child nodes per parent node

d = depth of the shortest path

FRET Graph Representation

Natural Chlorophylls in Et2O Solvent

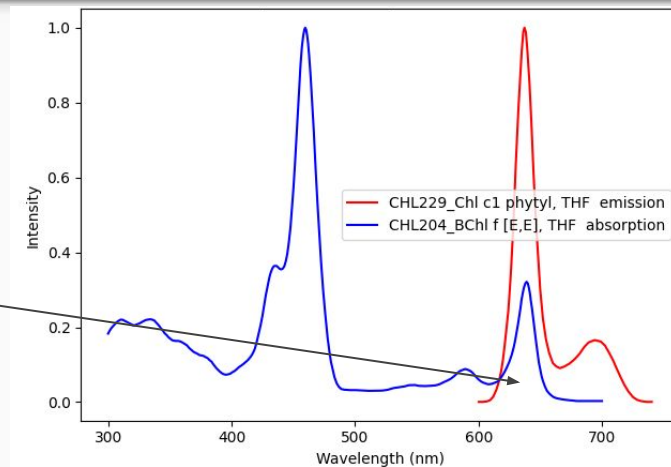


Nodes: individual chlorophyll variants

Edges: FRET Efficiency

Edge direction: emission → absorption

Spectral Overlap,
 J_{rel}



Edge weight:

- $-\log(E)$
- Additive for A^*
- Represents most probable energy transfer path

$$E = \frac{1}{1 + \left(\frac{r}{R_0}\right)^6}$$

$$R_0^6 \propto J \propto J_{rel}$$

$$E = \frac{1}{1 + \left(\frac{C}{J_{rel}}\right)}$$

A* Implementation

A* Applied

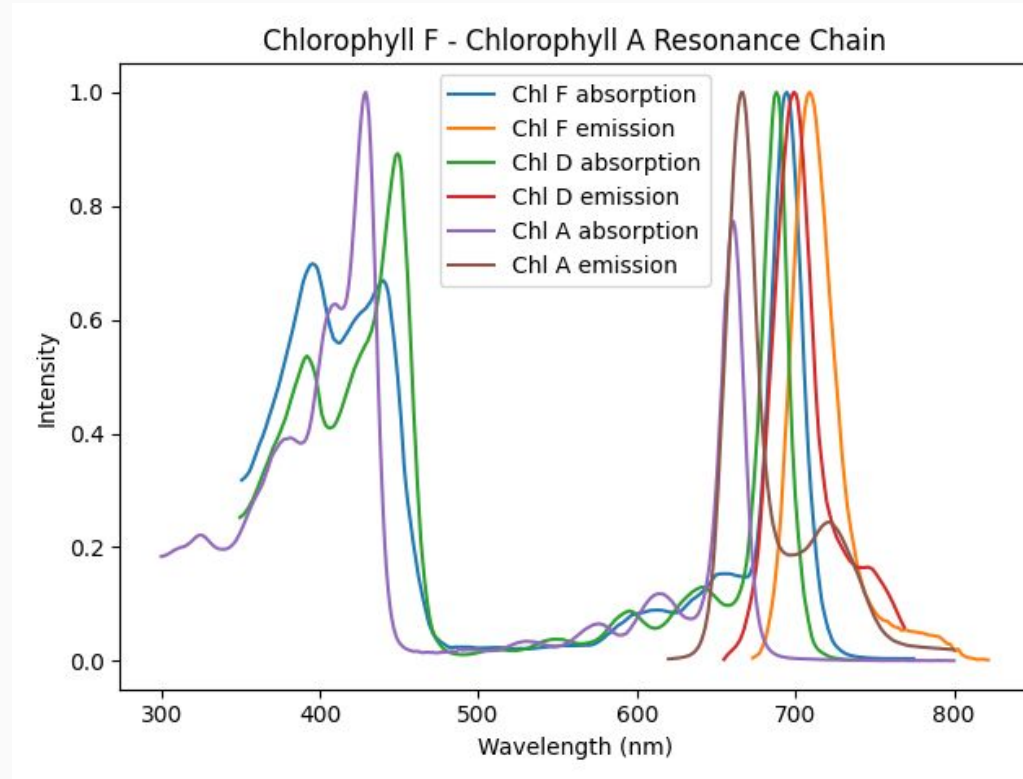
- On Natural Chlorophyll Et2O graph
- Target:
 - Chlorophyll A
 - Activates electron transport chain
- Source:
 - Chlorophyll F
 - Results in maximum attenuation
 - Expands spectral absorption range of photosynthetic light complex

Shortest Path:

Chlorophyll F → Chlorophyll D → Chlorophyll A

Broader Applications

- Computationally search for synthetic accessory pigments for photosynthesis
 - Expand spectral absorption range
 - Improve efficiency of existing pathways



Limitations

- Heuristic Function

- Given dataset including both quantum yield and spectral information
 - Could calculate actual fret efficiency
 - Integrals are computationally expensive
 - Setting $J=1$ in the fret equation provides a best case efficiency

$$E = \frac{1}{1 + \left(\frac{C'}{Q_{yield} \cdot J_{rel}} \right)}$$

$$h(n) = \frac{1}{1 - \frac{C'}{Q_{yield}}}$$

**Thank
you!!**